

POTPRESS · DESIGN DECISIONS REGISTER

PotPress design decisions register

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	PPR-DEC-001	Version	0.5
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/potpress		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-03	Amish Chadha	Amish decided the four guard fixes (2026-10-03); moved to decisions made; value engineering at USD 1,169
0.4	2026-10-02	Amish Chadha	Approved follow-ups carried out: four guard changes found by the ISO 13857 desk check added as open decisions; handle stroke added to the items to confirm; value engineering at USD 1,149
0.3	2026-10-02	Amish Chadha	Amish approved the recommendations for all five open decisions (2026-10-02); moved to decisions made
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-09-30	Amish Chadha	Register opened; open decisions moved out of the build plan

PotPress design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

None. The four guard fixes found by the ISO 13857 desk check were decided by Amish on 2026-10-03 and are in the table of decisions made. The desk check, rerun with them in the model, finds every opening meets ISO 13857 Table 4 on paper (PPR-CAL-001 v0.10, section 15); it still needs a competent person's signature, which is a hold point already decided on 2026-10-02.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	Buy the bottle jack first	Its base, release screw and pump socket set the locating blocks, the interlock coupling and the notch angle	PPR-DDR-004, Q4
2	Mold cavity finish that releases the liner cleanly	Sets the hand finishing of both molds	PPR-DDR-004
3	The pump handle's stroke on the jack bought, and its play in the socket	The slot and shield leave 25 mm beyond each end of the modelled stroke and 5 mm each side; a longer stroke or more play shortens those gaps	PPR-CAL-001, section 15

■ Value engineering

Value-engineering target: USD 1,060 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 1,169 (USD 109 over the target). Main cost drivers and savings worth trying:

- The press alone is USD 1,086 and the QC rack USD 83. The USD 1,060 target was set on 2026-09-27 to cover the whole BOM, QC rack included, so the like-for-like gap is USD 109.
- The fixed inner shield behind the pump slot (decided 2026-10-02) added USD 6. The four guard fixes (decided 2026-10-03, cost accepted by Amish) added USD 20 against the USD 15 estimated: finer mesh USD 8 (items 16 and 20), small-hole plates USD 2, top beam cover plates USD 7 (longer than estimated, to close the upright channel tops too), crank bracket covers USD 4. They are a safety requirement, so savings are sought elsewhere.
- Making the design constructable added USD 92 (PPR-DDR-004): the steel base plate under the female mold, the thicker 450 mm mold flanges, dowels and bushes (items 8 and 9, USD 44), floor anchors, the bolted jack plate and heavier feet (item 1, USD 22), the adapter disc and nut box (item 10, USD 9) and interlock details (item 21, USD 7).
- The fixed guards, front gate and interlock added USD 67 earlier (PPR-DDR-003); they are a safety requirement, so savings are sought elsewhere.
- Savings worth trying: cheaper dowels and bushes, and plywood QC rack shelves cut from offcuts.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items and recommendations	Amish: "i accept all your recommendations, go with them across all repos."	PPR-DDR-001, PPR-DDR-002
2026-09-27	Guarded version with mesh guards and an interlocked front gate; budget USD 1,060	Amish	PPR-DDR-003
2026-09-30	Design for construction: open-topped male mold, dowel location, M16 joints with spacer tubes, captive lead screw nut, interlock design and other changes that make the press buildable	Amish: "i accept your recommended changes on design that are currently being sent across for my approval"	PPR-DDR-004
2026-10-02	Pump slot: keep the 30 mm slot, approved only with a fixed inner shield or tunnel behind the slot that keeps the platen and molds out of arm's reach, sized in the ISO 13857 check (item 5)	Amish: "i approve your recommendations for all 555 open decisions."	PPR-DDR-004, Q2; PPR-DDR-003
2026-10-02	Pump handle: the 158 mm the handle stands outside the guard is operating space, like a door swing, marked on the floor layout; the 1.0 m width limit of R11 applies to the guarded box	Amish: "i approve your recommendations for all 555 open decisions."	PPR-DDR-004, Q3
2026-10-02	Demolding: keep the tipping pins for the prototype; the potters at the first partner factory confirm the method, and if they object the pot is lifted out in its liner instead	Amish: "i approve your recommendations for all 555 open decisions."	PPR-DDR-004, Q5
2026-10-02	Pressing force: every structural check stays at the full 20 t jack; the working force is a process setting found in pressing trials with a partner factory, starting near 2 t and stepping up	Amish: "i approve your recommendations for all 555 open decisions."	PPR-DDR-001, item 9 (as cited in the register); PPR-CAL-001, section 3
2026-10-02	Guard openings: the ISO 13857 desk check is done now rather than at TRL 4, signed by a competent person, and is a hold point before any force above hand pressure	Amish: "i approve your recommendations for all 555 open decisions."	PPR-DDR-003; PPR-DDR-004, Q2
2026-10-03	Front and roof mesh: 6.35 mm (1/4 in) welded mesh, 0.9 mm wire, on the front strips, lower front panel, front gate and roof (option (a)); the side and rear guards keep the 12.7 mm mesh	Amish: "PotPress - move forward with the four guard fixes. I accept the cost."	PPR-CAL-001, section 15; BOM items 16 and 20
2026-10-03	Release shaft and pin cable openings: bolted plates behind the front right strip with a 20 mm hole round the 12 mm shaft and a 10 mm grommeted hole round the 6 mm cable (option (a))	Amish: "PotPress - move forward with the four guard fixes. I accept the cost."	PPR-CAL-001, section 15; BOM item 16
2026-10-03	Top beam gap: 3 mm steel cover plates bolted to the top flanges (option (a)); as modelled, one plate each side from the crank bracket to the beam end, which also closes the open tops of the upright channels	Amish: "PotPress - move forward with the four guard fixes. I accept the cost."	PPR-CAL-001, section 15; BOM item 16
2026-10-03			

Date	Decision	Decided by	Record
	Crank bracket: 2 mm sheet covers on its open front and back (option (a)); costed with the guards, BOM item 16	Amish: "PotPress - move forward with the four guard fixes. I accept the cost."	PPR-CAL-001, section 15; BOM item 16